

CSAI 801 Artificial Intelligence & Machine Learning

Machine Learning Project Proposal

Understanding the Macroeconomic Signals

Behind the US Dollar Index (DXY)

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Motivation

Exchange rates play a significant role in global economic stability. The strength of the US dollar affects international trade, investment flows, and inflation levels in economies connected to USD transactions such as Canada. While many studies attempt to predict exchange rate movements, market reactions to economic indicators are rarely immediate and usually occur with delay.

Instead of focusing only on predicting the next value of DXY, our project analyzes which macroeconomic indicators contain useful information about future movements of the dollar and how early that information becomes informative.

Dataset

Target Variable: Monthly DXY values from Yahoo Finance / Investing.com

Macroeconomic Indicators from FRED:

All variables will be aligned to a monthly frequency. The selected indicators serve as a starting set and may be adjusted depending on data availability and consistency after preprocessing.

- | | |
|--------------------------------|------------------------------|
| ▪ Initial Jobless Claims | ▪ Housing Price Index |
| ▪ Federal Funds Rate, GDP | ▪ Existing Home Sales |
| ▪ Unemployment Rate | ▪ CPI (Food & Energy) |
| ▪ Employment Level | ▪ Core CPI, Commodity Prices |
| ▪ Average Weekly Working Hours | ▪ USD/CAD Exchange Rate |
| ▪ Manufacturing Activity Index | |

Data sources:

- Federal Reserve Bank of St. Louis, "Federal Reserve Economic Data (FRED)," [Online]. Available: <https://fred.stlouisfed.org/>
- Yahoo Finance, "U.S. Dollar Index (DXY) historical data," [Online]. Available: <https://finance.yahoo.com/quote/DX-Y.NYB/>

Related Work

Previous research in exchange rate forecasting showed that classical macroeconomic models often fail to outperform simple random walk predictions (Meese & Rogoff, 1983) [1]. Machine learning models later demonstrated improved predictive performance (Zhang et al., 2019; Baek & Kim, 2020) [2].

However, most studies evaluate success mainly through prediction accuracy. Recent econometric perspectives suggest machine learning can also be used for understanding relationships within economic data (Mullainathan & Spiess, 2017) [3]. This project follows that direction by analyzing temporal predictive information rather than focusing only on accuracy. Machine learning techniques have also been applied to both daily and monthly exchange rate forecasting problems [4].

Intended Experiments \ Methodology

- 1- We will first align all variables to a monthly frequency and construct lagged versions of each macroeconomic indicator to capture delayed market reactions.
- 2- A baseline model will be created using a naïve assumption that the next value of the index is like the most recent observation.
- 3- We will then train several regression models including Linear Regression, regularized regression (Ridge/Lasso), and a tree-based model (Random Forest or Gradient Boosting).
- 4- After training, we will analyze feature importance and compare how the predictive contribution of each indicator changes when different time delays are used.

Evaluation

Performance will be measured using MAE and RMSE and compared with a naive baseline. The main objective is interpretability: identifying which indicators contain predictive information and when their effect appears.

Expected Outcome

The goal of this project is to analyze a set of macroeconomic indicators to identify which ones provide useful information about future movements in the US Dollar Index and how early their influence becomes noticeable. The outcome is not only a predictive model but also a clear explanation of which indicators tend to precede changes in the index.

References

- [1] R. A. Meese and K. Rogoff, "Empirical exchange rate models of the seventies: Do they fit out of sample?" *Journal of International Economics*, 1983. Available: [https://doi.org/10.1016/0022-1996\(83\)90017-X](https://doi.org/10.1016/0022-1996(83)90017-X)
- [2] Y. Baek and H. Y. Kim, "Modifying the random forest to improve the prediction of exchange rates," *Global Economic Review*, 2020. Available: <https://doi.org/10.1080/1226508X.2020.1741258>
- [3] S. Mullainathan and J. Spiess, "Machine learning: An applied econometric approach," *Journal of Economic Perspectives*, 2017. Available: <https://doi.org/10.1257/jep.31.2.87>
- [4] V. Plakandaras, P. Gogas, and T. Papadimitriou, "Forecasting daily and monthly exchange rates with machine learning techniques," *Journal of Forecasting*, 2019. Available: <https://doi.org/10.1002/for.2584>